NO TIME TO HACK (NTTH)

Progress, Comparison & Novelty Report

A Behavior-Driven Autonomous IoT Security Gateway with
Hybrid Intrusion Detection and Containerized Deception

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This document presents our project journey from the initial proposal to the current implementation,
compares our work against existing published research, and demonstrates the novelty of our approach.

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Section 1: Initial Idea vs. Where We Are Now

1.1 The Initial Proposal (What We Pitched)

When we first proposed this project, our idea was straightforward on paper — build a smart gateway that can watch over a home network and automatically protect IoT devices from attacks. We envisioned a system that could:

- Sit as an Access Point so all device traffic flows through us
- Detect common attacks like port scans and brute-force login attempts
- Use some form of machine learning to catch unusual behavior
- Redirect attackers to fake servers (honeypots) instead of real devices
- Block or isolate dangerous devices automatically
- Show everything on a dashboard in real time

Honestly, at the proposal stage, we had a conceptual understanding of each piece but hadn't figured out how they would all connect. The proposal described an "Observe–Analyze–Decide–Act" pipeline, but the actual architecture — the event bus, the agent stages, the NAT redirection logic — none of that existed yet. It was an idea, not a design.

1.2 Where We Are Right Now (Actual Implementation)

Fast forward to today, and we have a fully functional, deployed system that goes far beyond what we initially proposed. Here is an honest accounting of what is built and running:

Core Backend — 100% Complete

Component	Technology	Status	Evidence
API Server	Python FastAPI	DONE	12 REST endpoints, running on port 8000
Authentication	JWT (login/refresh)	DONE	Secure admin access verified
Database	PostgreSQL + SQLAlchemy	DONE	515+ threat events stored
Real-time Updates	WebSocket	DONE	Live push to dashboard clients

Configuration	Pydantic Settings + .env	DONE	All parameters configurable
Logging	Structured JSON	DONE	File + console output

Intrusion Detection System — 100% Complete

Component	How It Works	Status
Packet Sniffer	Scapy AsyncSniffer on network interface, captures every IP packet	DONE
Feature Extractor	Extracts src_ip, dst_port, TCP flags, protocol, packet length	DONE
Port Scan Detector	Sliding window: 4+ unique ports in 15 seconds → score 1.0	DONE
SYN Flood Detector	Sliding window: 30+ SYN packets per second → score 1.0	DONE
Brute Force Detector	Sliding window: 3+ auth-port hits in 120 seconds → score 1.0	DONE
Isolation Forest ML	200 trees, trains on 500 baseline packets, scores anomalies 0–1	DONE
Risk Calculator	Hybrid: 0.6 × rule_score + 0.4 × ml_score	DONE

Autonomous Agent Pipeline — 100% Complete

Agent	Responsibility	Status
Event Bus	Async pub/sub queue (5000 capacity) connecting all stages	DONE
Threat Agent	Runs IDS + ML scoring, enriches with GeoIP, publishes threats	DONE
Decision Agent	Protocol-aware action selection: block / honeypot / rate-limit	DONE
Enforcement Agent	Applies nftables kernel firewall rules, registers redirect contexts	DONE
Reporting Agent	Persists events to PostgreSQL, broadcasts via WebSocket	DONE

Firewall & Honeypot — 100% Complete

Component	Details	Status
nftables Manager	Block, rate-limit, and NAT redirect rules via kernel firewall	DONE
Rule Tracker	Deduplication + DB tracking + auto-expiry after TTL	DONE

Cowrie SSH Honeypot	Docker container, port 30022, captures commands and credentials	DONE
HTTP Honeypot	Built-in Python server on port 8888, logs attack probes	DONE
Cowrie Log Watcher	Real-time JSON log tail with session correlation	DONE

Flutter Dashboard — 100% Complete

Screen	Features	Status
Login Screen	JWT authentication, configurable server URL	DONE
Dashboard Overview	Summary cards with key statistics	DONE
Threat Map	Live/Recent badges, risk score chips, real-time updates	DONE
Firewall Rules	Active/History tabs, NEW/EXPIRED badges	DONE
Honeypot Sessions	Terminal-style command display, credential capture	DONE
Network Topology	ARP-based device discovery and risk visualization	DONE

> **Honest Note to Professor:** The entire system implementation — backend, IDS, agents, firewall, honeypots, and dashboard — is 100% built and has been tested with real attacks from an Android device running Termux on the same LAN. This is not a simulation. We have 515+ real threat events in our database from live testing.

1.3 How the Initial Idea Evolved

Aspect	Initial Proposal	Current Reality	Change
Architecture	Vague "layered gateway"	4-stage agent pipeline with async event bus	Major upgrade
Detection	"Some ML-based detection"	Hybrid: 3 rule detectors + Isolation Forest, weighted 0.6/0.4	Much deeper
Enforcement	"Block suspicious devices"	5-tier graded response: allow → log → rate-limit → honeypot → block	Significantly richer
Honeypot	"Deploy honeypot"	Flow-aware NAT redirect per attacker-victim-port flow + Cowrie + HTTP honeypot	Novel contribution
Firewall	"iptables rules"	nftables kernel firewall with rule tracking, dedup, and auto-cleanup	Production-grade
Dashboard	"Web interface"	Cross-platform Flutter app (Web + Android) with WebSocket live updates	Beyond expectation

Database	Not mentioned	PostgreSQL with 4 tables: threats, rules, sessions, devices	Added
GeoIP	Not mentioned	MaxMind GeoLite2 enrichment (country, city, ASN, lat/lon)	Added

Section 2: Progress From Final Goal to Current State

2.1 Overall Progress Visualization

System Implementation

100% — All 47 components built

Research Paper

75% — Structure complete, results pending

Experimental Validation

demos done,
formal metrics

Wireless Monitoring (AR9271)

d for

Combined Overall Progress

~65% Complete

2.2 What's Done vs. What Remains

Area	Status	Items Done	Items Remaining
Backend + API	100%	8 of 8	0
IDS + ML	100%	10 of 10	0
Agent Pipeline	100%	5 of 5	0
Firewall	100%	3 of 3	0
Honeypot System	100%	5 of 5	0
Flutter Dashboard	100%	9 of 9	0
Documentation	100%	8 of 8	0
Formal Experiments	15%	Live demos	Controlled metrics collection, Snort/Suricata comparison, CICIDS2017 validation

Research Paper	75%	Full structure + content	Insert verified results, IEEE formatting
Wireless (AR9271)	0%	—	Monitor mode, probe tracking, deauth detection

2.3 Remaining Work — Prioritized Plan

#	Task	Priority	Time Est.	Why It Matters
1	Run 180 controlled attack experiments (50 port scan + 50 brute force + 30 SYN flood + 50 HTTP probe)	CRITICAL	2 days	Provides detection rate and FP rate numbers
2	Measure end-to-end latency T1→T4 for 200 packets	CRITICAL	1 day	Our strongest novelty metric (~127ms)
3	Install Snort + Suricata, run same attacks, compare	CRITICAL	2 days	Proves we outperform existing IDS
4	Ablation study: test 11 weight configurations	CRITICAL	1 day	Validates the 0.6/0.4 weight choice
5	Multi-model comparison (RF, SVM, KNN, DT vs IF)	CRITICAL	2 days	Justifies Isolation Forest selection
6	Validate on CICIDS2017 benchmark dataset	HIGH	2 days	Cross-validates on 2.8M record standard
7	Finalize IEEE-format research paper	CRITICAL	5 days	The final deliverable
8	AR9271 wireless monitoring module	MEDIUM	3 days	Adds wireless novelty dimension

> **Honest Assessment:** The system implementation is complete and tested with real attacks. What remains is primarily the *scientific validation* — running controlled experiments, collecting formal metrics, and documenting them in the paper. The engineering is done; the science documentation is what we need to finish.

— End of Part 1 — Continued in Part 2: Detailed Research Paper Comparison —